OFF-GRID VEHICLE TRACKER

USER MANUAL, FOR EVERYDAY DRIVERS

Two radios, two dashboard displays, and nothing else. Each vehicle always knows where the other one is: distance, direction, speed, and short messages, with no cell phone, no cell tower, and no internet anywhere in the system.



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1. What this system is

This kit lets two vehicles keep track of each other anywhere, including places with no phone signal at all: deserts, mountains, long rural highways, another country. Each vehicle carries a small green **radio** and a dashboard **display**. The radios talk directly to each other over the air, like walkie-talkies that never need pressing. Every few seconds, each radio quietly tells the other one "here is where I am." Your display turns that into an arrow, a distance, and plain words.

> **In plain words:** it works like a compass that points at your friend instead of north. No accounts, no apps, no monthly fees, no coverage maps. If both boxes are switched on and charged, it works.

The two radios have names, written on their cases: **SF** and **RG**. Your display shows your partner's name at the top, so you always know who the arrow is pointing at.

2. What is in the kit

Item	What it does	What to check
Radio "SF" (green case)	Vehicle 1's transmitter. It finds its own position from GPS satellites and shares it over the air.	Confirm the antenna is screwed on, the battery is charged, and the small screen lights up.
Radio "RG" (green case)	Vehicle 2's transmitter. It does the same job as SF.	Apply the same checks as for SF.
Display 1	The dashboard screen that rides with SF. It shows where RG is.	Confirm it powers on and shows "link OK" at the bottom once its radio is on.
Display 2	The dashboard screen that rides with RG. It shows where SF is.	Apply the same checks as for Display 1.
Antennas (black sticks)	The radios' voice and ears. Nothing works without them.	Make sure each is hand-tight on its round connector. Never transmit without one.
USB-C cables and chargers	These charge the radios and power the displays.	Charge the radios after every trip.

> **Important pairing:** each display belongs to one radio. Display 1 works with SF. Display 2 works with RG. Keep each pair together in the same vehicle.

3. Before every trip: the two-minute start-up

PRE-FLIGHT CHECKLIST

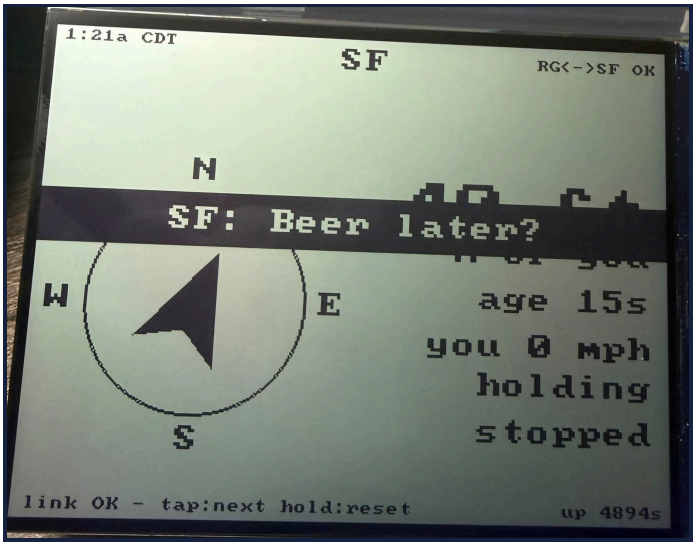
1. **Charge check.** Make sure both radios are above 20 percent. A radio that dies mid-trip can forget its settings, and then the two radios stop understanding each other.
2. **Antennas on.** Screw an antenna onto each radio, hand-tight.
3. **The magic number.** On each radio, tap the button to reach the screen labeled "LoRa" and confirm it reads **912.875**. Both radios must show this exact number. It is the private channel the two radios share. If either shows a different number, see Section 8.
4. **Power on together.** In each vehicle, switch on the radio and the display at the same time, like turning the ignition key. The display finds its radio during the radio's first minute. The display footer then reads `link OK`.
5. **Sky view.** Position updates need GPS. Give each radio a view of the sky (dashboard, window mount). Indoors, in a garage, positions will honestly show as missing until you drive out.

Within a minute or two of both vehicles being on, each display shows the partner's name, an arrow, and a distance. You are ready to drive.

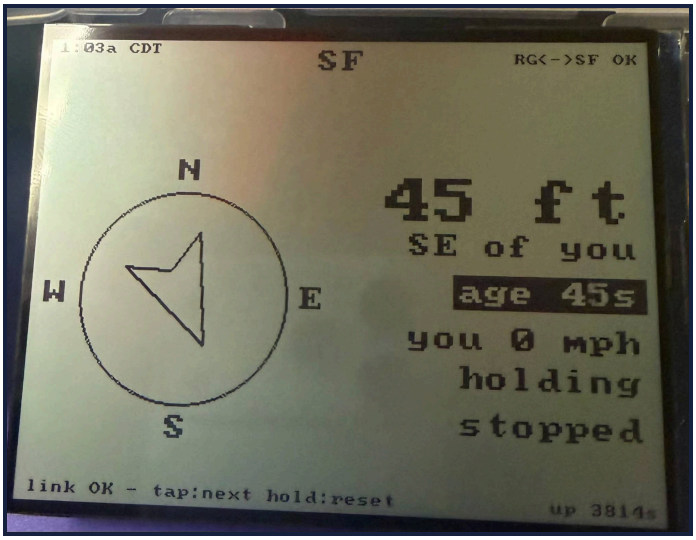
4. Reading your screen

The display has three pages. A short press of its button (`tap: next`) moves to the next page.

Page 1: The compass (your driving page)



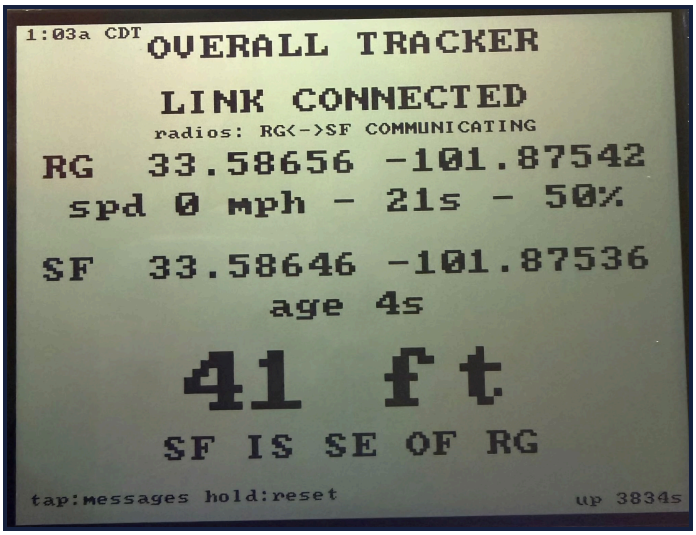
compass page, with a message arriving



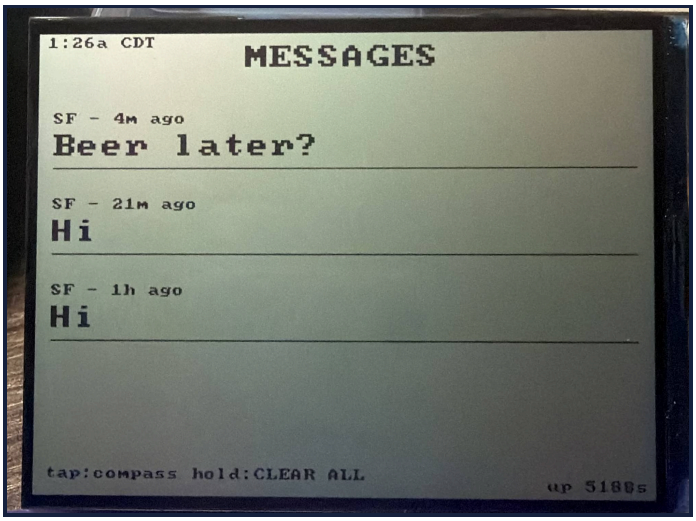
compass page, data getting old (hollow arrow)

What you see	What it means
Name at the top (SF or RG)	This is who the arrow points at: your partner vehicle.
The dark arrow in the circle	This is the direction to your partner. A solid arrow means the position is fresh. A hollow outline arrow means the position is an estimate, because no update has arrived for half a minute, so trust it less.
Big number (feet or miles)	This is the straight-line distance to your partner.
"SE of you" or "at 2 o'clock"	This is the direction in words. While you drive, it uses clock positions relative to your hood. While you are stopped, it uses compass words.
"age 15s"	This is how old the partner's information is, in seconds. Small is good. If it turns into a dark highlighted box, the data is getting stale.
"you 0 mph"	This is your own speed.
"closing 12 mph" / "apart 8 mph" / "holding"	This tells you whether the gap between the vehicles is shrinking, growing, or steady.
"RG<->SF OK" (top right)	This means the radios are actively hearing each other. If the tag disappears, they have gone quiet.
"link OK" (bottom)	This means the display is connected to its own radio in your vehicle.

Page 2: The status page (the numbers page)



status page: positions, ages, battery, distance



messages page with history

This page lists both vehicles by name with their exact coordinates, speeds, data ages, and each radio's battery percentage. The line "radios: RG<->SF COMMUNICATING" confirms the two radios are in contact. The big number is the distance between the two vehicles, and the sentence under it, such as "SF IS SE OF RG", says the same thing in words.

Page 3: Messages

The last few messages you received, newest first, each with the sender's name and how long ago it arrived. Holding the button on this page (`hold: CLEAR ALL`) erases the list. Holding the button on any other page resets the tracking data instead, which is harmless: fresh data rebuilds within seconds.

5. Sending and receiving messages

Messages are **sent from the radio** (the small green box) and **appear on the partner's display**. The radio stores 21 ready-made messages so nobody types while driving. Press the radio's button to open the list, keep pressing to scroll, and long-press to send the highlighted one.

Group	Messages
Quick replies	OK · Yes · No · Hello
Checking in	How are you? · I am good
Safety	Help! · Are you hurt?
Position	Send position
Driving	On my way · Wait there · I am here · Follow me · Slow down · Stop · Turn left · Turn right · Turn back
Social and status	Beer later? · Low battery · Going home

When a message arrives, it appears as a dark banner across the partner's display for about eight seconds, for example **"SF: Beer later?"**, and is also saved on the Messages page. Only the receiving vehicle gets the banner. Your own display stays quiet when you send, which is how you know the banner you see is always news from the other car.

> **In plain words:** every question in the list has its answer in the list. "Are you hurt?" can be answered with "I am good" or "Help!". "Beer later?" can be answered with "Yes". The list is a conversation kit.

6. Asking "where are you right now?"

Positions update automatically: every ten seconds while a vehicle is moving, every minute while it is parked. If you want the partner's position immediately, press the **BOOT button on your display**. Your radio asks the partner's radio directly, and the partner's radio answers within a few seconds. You will see the partner's "age" number drop to nearly zero. This is useful when the partner has been parked and quiet and you want to confirm where they are before setting off.

7. Warnings, and what to do about them

On the screen	What it means	What to do
RADIO LINK LOST	The display cannot reach its own radio in your vehicle.	Check the radio is on and within a few feet. Power the radio and display off, then on together.
SF DATA STALE (or RG)	No position has arrived from the partner for over two minutes.	This is usually distance or terrain at work. The arrow hollows and the age counts up honestly, and it heals itself when you get closer.
RG BATTERY 15% (or SF)	The partner's radio battery is nearly empty.	Tell them to plug it in. A dead radio disappears from your screen entirely.
Hollow arrow, highlighted age	The position shown is an estimate, not a fresh measurement.	Treat the arrow as approximate until it turns solid again.
CHANGE COURSE >>	You have been driving away from your partner for a while.	Follow the chevrons if your goal is to rejoin them.

8. When something seems wrong

"The other vehicle is not on my screen at all."

1. Is the partner's radio switched on, with its screen lit? A dark screen means a dead battery.
2. Does each radio's LoRa screen read **912.875**? If one shows a different number, that radio lost its settings, usually after its battery ran completely flat. It needs the recovery procedure in the technical manual, a five-minute fix with a laptop.

3. Power-cycle both radios (off, count to five, on). They announce themselves on boot and usually reappear on each other's screens within a minute.

"The distance looks wrong."

Distance is measured between two GPS positions. Indoors or parked by a building, each GPS can be off by a few dozen feet, so two parked vehicles may read 40 feet apart while sitting side by side. This tightens up outdoors and while moving. If distance is wildly wrong (miles apart while parked together), one radio is holding an old position: press the display's BOOT button to request a fresh one.

"The names show letters and numbers instead of SF or RG."

The display learns names from the radios' introductions. After a reset it may briefly show a serial-style name like "bb5c". It corrects itself automatically; power-cycling the partner radio speeds it up.

9. Taking care of the equipment

- **Charge after every trip.** A radio left switched on runs its battery flat in one to two days.
- **Switch radios off when stored.** Off, they hold charge for months. A battery that runs completely flat can erase the radio's settings.
- **Never transmit without an antenna.** Operating a radio with the antenna removed can damage it permanently.
- **The printed cases** protect the screens and keep every port reachable. If a case is ever opened, use only the original short screws: longer screws press on the glass.
- **Sun and heat.** The dashboard display is sunlight-readable by design and produces almost no heat. The radios prefer shade like any battery device.

10. One-page quick reference

I want to...	Do this
Start the system	Attach the antennas, check the charge, power the radio and display together in each vehicle, and confirm "link OK".
See where my partner is	Read the compass page: the arrow, the distance, the direction in words, and the data age.
See exact numbers and batteries	Tap the display button to reach the status page.
Send a message	On the radio, press to open the list, press again to scroll, and long-press to send.
Read past messages	Tap the display button to reach the Messages page.
Erase past messages	Hold the display button while on the Messages page.
Get partner's position right now	Press the BOOT button on the display.
Fix "partner missing"	Check the partner's power, confirm both radios read 912.875, and power-cycle both.
Store the kit	Charge everything, then switch the radios off.

11. How it works, in five ideas (no math, promise)

You do not need this chapter to use the system. It is here because the system behaves more sensibly once you know what is going on inside, and because every rule in this manual comes from one of these five ideas.

> **Idea 1: agreement is the connection.** The radios never "pair" or "log in" to each other. They simply both follow the same three rules: the same frequency, the same way of speaking, and the same secret code. As long as all three match, they hear each other. Every time this system has ever gone silent, one radio had quietly forgotten one of the three. That is why the pre-flight check exists.

> **Idea 2: the name is the frequency.** Our radios turn the channel name "Group9" into a frequency using arithmetic, and the answer is 912.875. Both radios do the same arithmetic on the same name, so they always land on the same spot on the dial without anyone tuning anything. That is why checking for 912.875 on the little screen is such a powerful test: if the number is right, the radio remembered its settings. If it is wrong, the radio forgot them, usually because its battery ran completely flat.

> **Idea 3: the signal whispers below the noise.** These radios use a trick called LoRa, short for Long Range. Instead of shouting fast, they speak slowly in long sweeping tones that a receiver can pick out even when the signal is quieter than the background static. That is how two lunchbox-sized radios cover miles of open country on less power than a flashlight. The price of that trick is speed: each little report takes about half a second of airtime, which is why positions update every ten seconds rather than constantly.

> **Idea 4: "online" is just a stopwatch.** No wire connects the vehicles, so your display cannot truly know the partner is there. What it knows is when it last heard from them. Everything on the screen that looks like a connection status, the OK tag, the age number, the stale warning, is a stopwatch since the last message. Small age: fresh news. Growing age: the partner has gone quiet, maybe behind a hill, maybe parked with the radio off. The screen never pretends; the hollow arrow and the highlighted age are it telling you honestly that its news is old.

> **Idea 5: airtime is a shared budget.** Only one radio can speak on the channel at a time, like people sharing one walkie-talkie frequency. Every position report spends a slice of that shared airtime. The system is tuned so both vehicles reporting every ten seconds still leaves most of the air free for messages and requests. That is why it does not update every single second: it could, briefly, but the channel would jam and the radios would start skipping reports.

12. Questions people actually ask

Can other people see where we are?

No. Everything the radios send is scrambled with a secret code that only our two radios hold. Someone with a scanner could tell that two radios are talking somewhere nearby, but not what they said and not where you are.

Does it need cell coverage or wifi anywhere?

No. The radios talk straight to each other through the air, and they learn their own positions from GPS satellites, which are free to listen to everywhere on Earth. Nothing in the kit ever touches the internet.

How far apart can the vehicles get?

The design comfortably covers 2 to 3 miles (3 to 5 km) with hills and buildings in the way, and farther on flat open ground. When you drive out of range the screen does not lie: the age counts up, the arrow hollows, and the STALE warning appears. Drive back toward the last known direction and it heals on its own.

Is there a monthly fee or subscription?

No. The radios use a license-free public band, the same kind of airspace as garage door openers and baby monitors, just used more cleverly.

What happens if one battery dies mid-trip?

The other vehicle's screen simply shows the partner's age climbing and then the stale warning, with the last known position still displayed. Recharge and switch back on, and the pair finds each other again within a minute. Also charge it soon: a battery that runs completely flat can erase the radio's settings (Section 8).

Can more vehicles join?

The radios are built on technology that supports many nodes, so a third vehicle is possible with one more radio and display and a small settings change. The current kit is tuned for exactly two.

Why does my parked distance jiggle between 15 and 45 feet?

Each vehicle's GPS guess wanders by a few dozen feet when standing still, and the distance is the gap between two wandering guesses. It is normal, and it settles once you move.

Appendix: words and abbreviations, in plain English

Word	What it means here
Antenna	The black stick on the radio. It is the radio's mouth and ears, and you should never transmit without it.
Age	The number of seconds since the partner's last report arrived. It is the screen's honesty meter.
BLE (Bluetooth Low Energy)	The short-range wireless link between the display and its own radio inside your vehicle. This is the "link OK" at the bottom of the screen.
BOOT button	The small button on the display. Pressing it asks the partner's radio for a fresh position right now.
Channel	The private lane of airspace our two radios share, named Group9.
Encryption	Scrambling messages with a secret code so only our radios can read them.
GPS (Global Positioning System)	Satellites that let each radio work out its own location. Listening is free and works everywhere with a sky view.
Hollow arrow	The compass arrow drawn as an outline: the position shown is an estimate because the news is more than half a minute old.
LoRa (Long Range)	The slow-but-mighty radio technique that carries the signal for miles on tiny power (Idea 3).
Meshtastic	The free, open radio software the green radios run. It handles positions, messages, and the secret code.
MHz (megahertz)	The unit for a spot on the radio dial. Our spot is 912.875 MHz.
Power cycle	Switch the device off, count to five, and switch it on again. It is the first move for almost any electronic confusion.
Pre-flight check	The two-minute Section 3 routine before every trip. Named after the checklist pilots run before takeoff.
SF and RG	The names of the two radios, printed on their cases. Your screen shows your partner's name.
Stale	Old enough that you should not fully trust it. Appears when no report has arrived for two minutes.
USB-C	The oval charging plug, the same one most modern phones use.
912.875	The magic number: the exact frequency both radios must show on their LoRa screen. If both show it, they can hear each other.

Off-Grid Vehicle Tracker · Group 9, Texas Tech University ECE Capstone · The radios run Meshtastic open-source firmware (meshtastic.org); the displays run software written by Group 9. For engineering details, derivations, and full references, see the companion Technical Manual.